# NC Property Tax Demo — Methodology Notes

Per-property comparison under HB 1089: data sources, benchmark construction, and calculation contract

Version 1.0 · September 2026 · North Carolina fiscal years 2016–17 through 2025–26

## 1. Purpose

This document describes how the per-property tax comparison on the site is constructed. For a selected residential parcel, the page compares two ways of taxing the same assessed value at the county level:

- **What you paid** — the county's actual FY2025–26 county rate applied to your parcel's assessed value.
- **What you could have paid** — the rate implied by a benchmark levy that grows the FY2016–17 county-wide levy each year by inflation (U.S. CPI–U) plus county population growth, capped at the actual FY2025–26 levy.

The comparison holds the parcel's assessed value constant and changes only the county tax rate, so the result isolates the effect of *rate growth* relative to a minimal, status-quo benchmark.

## 2. Data sources

| Input | Source | Used for |
|---|---|---|
| FY2016–17 county-wide levy `l_2016` | NCDOR LG04 (county/municipal levy and valuation report) | Benchmark baseline |
| FY2025–26 county-wide levy `l_actual` | NCDOR LG04 | Scenario cap; reconciliation |
| FY2025–26 taxable base `x` | NCDOR LG04 (total assessed valuation) | Implying the scenario rate |
| FY2025–26 county rate `r_actual` | NCDOR county and municipal tax rate schedule | "What you paid" |
| FY2016–17 county rate `r_2016` | NCDOR 2016–17 tax rate schedule | Documentation/reconciliation |
| Inflation | BLS national CPI-U annual averages (all urban consumers) | Benchmark growth, one component |

| Input | Source | Used for |
|---|---|---|
| Population | U.S. Census Bureau Population Estimates Program (vintage estimates) | Benchmark growth, one component |
| Parcel assessed value `V` | NC OneMap parcel point service | "What you paid" per property |

## 3. Benchmark construction

Nine growth transitions compound the FY2016–17 baseline levy to the FY2025–26 benchmark ceiling.

```
B_2016 = l_2016
B_t    = B_(t−1) × (1 + inflation_t + population_t)    for t = 2017, …, 2025
```

where `inflation_t` is the CPI-U annual-average change from year t−1 to t, and `population_t` is the county's annual resident population change from year t−1 to t. Each transition year's growth is the sum of those two components.

### 3.1 Scenario levy (capped) and scenario rate

```
L_scenario = min(l_actual, B_2025)
r_scenario   = 100 × L_scenario / x
```

The benchmark never prescribes an increase: if the county's actual levy is already *below* the inflation-plus-population ceiling (as in Mecklenburg), the actual levy is retained. The scenario rate is that capped levy expressed per `$100` of current taxable base.

### 3.2 Per-property comparison

```
tax_actual    = V × r_actual / 100
tax_scenario  = V × r_scenario / 100
difference   = tax_actual − tax_scenario
```

`V` is the parcel's assessed value from NC OneMap (ordinary residential parcels only).

## 4. Population & inflation detail

### 4.1 Population

County annual resident population uses the Census Bureau's vintage estimates, mapped so each transition year t uses the change between the t−1 and t population estimates. Because the Census retroactively revises population after each decennial benchmark, two vintages are used consistently:

- 2016→2019 transitions: the 2010–2020 vintage series (`co-est2020-alldata.csv`), i.e. the 2010-based estimates.
- 2020→2025 transitions: the 2020–2025 vintage series (`co-est2025-alldata.csv`), i.e. the 2025-based (post-benchmark) estimates — including the revised 2020 value used to bridge the two vintages.

This is the only place the two source vintages differ numerically from a single-series projection; the methodology is identical for every county.

### 4.2 Inflation

Inflation is the national CPI-U annual average (BLS, all items, not seasonally adjusted). The same series is used for every county. The annual averages underlying the transitions are:

| Year | CPI-U annual avg. | Transition into year | Inflation change |
|---|---|---|---|
| 2016 | 240.007 | — | — |
| 2017 | 245.120 | 2017 | 0.021304 |
| 2018 | 251.107 | 2018 | 0.024425 |
| 2019 | 255.657 | 2019 | 0.018120 |
| 2020 | 258.811 | 2020 | 0.012337 |
| 2021 | 270.970 | 2021 | 0.046980 |
| 2022 | 292.655 | 2022 | 0.080027 |
| 2023 | 304.702 | 2023 | 0.041165 |
| 2024 | 313.689 | 2024 | 0.029494 |
| 2025 | 321.943 | 2025 | 0.026313 |

## 5. Reconciliation & data notes

- **Rate/base/levy reconciliation.** Each county file records `rate_reconciliation_pct = 100 × (x × r_actual / 100 − l_actual) / l_actual`, the gap between the reported levy and the rate × base. Most counties reconcile within a few tenths of a percent; a small number differ by more (typically from mid-year reappraisals or tax-list year offsets in the underlying NCDOR records). The reconciliation is diagnostic metadata and does not affect the output, since the scenario rate is computed directly from the reported levy and base.
- **Parcel values.** NC OneMap parcel data is a live snapshot and may predate the 2025 tax year. The 2025 county rate and levy figures are the official FY2025–26 records.

- **Residential only.** Relief/exemption cases and non-residential parcels are out of scope.
- **Assumptions.** The ceiling compounds from the 2016 baseline; a year below the ceiling does not ratchet it down; there is no separate new-construction allowance or other override.

## 6. Interpretation

A county at or below the benchmark (actual levy ≤ ceiling) means current county property tax revenue has not grown faster than inflation-plus-population since 2016; the per-property difference is effectively zero and the site says so explicitly. A county above the benchmark shows the dollars a property at that assessed value would have saved had the county tied its levy growth to the benchmark rather than its actual rate path.

---

Repo: github.com/mkale-jlf/property-tax-demo · Calculation contract implemented in `calc.js` and covered by `test/calc.test.mjs` (17 tests).